TEST REPORT - TMTS (MOLD, S45C)

Member IMC Group

Header

Customer No.	9354
Product Line	4 - Milling
Industry	V2 - MOLDS & DIES - IP
Test Date (DD/MM/YYYY)	26/02/2026
Test Reason	1 - Productivity
Stage	3 - Completed
Test Result	1 - Positive
Branch	SEOUL

Machine Information

Type	1 - Milling Machine - Vertical
Brand & Model	DOOSAN MYNX 6500/50
Adaptation	BT50
Max. RPM	6,000

Cutting Condition

		Actual	Proposal 1	Proposal 2	Proposal 3	Proposal 4	Proposal 5
Manufacturer		224 - Pramet	1 - Iscar	1 - Iscar	1 - Iscar		
Tool		25A4R042B25-SAD11E-C	MDR D25-2-C25-AN12	MDR D25-2-C25-AN12	FFX4 ED25-4-060-C25-04		
Diameter (mm)		25	25	25	25		
No. of Flutes		4	2	2	4		
Insert		ADEX 11T308SR-HF2	MDR ANMU 120408PNTR	MDR ANMU 120408PNTR	FFX4 XNMU 040310T		
Carbide Grade		M8330	IC830	IC830	IC830		
Speed	Cutting Speed (m/min)	236	200	200	150		
	Spindle Speed (rpm)	**3006**	**2548**	**2548**	**1911**		
Feed	Feed per Tooth (mm/t)	0.18	0.2	0.2	0.84		
	Table Feed (mm/min)	**2164.6**	**1019.1**	**1019.1**	**6420.4**		
Width of Cut (mm)		25	25	25	25		
Depth of Cut (mm)		0.3	1	1.5	0.5		
MRR (Q, cm3/min)		**16.2**	**25.5**	**38.2**	**80.3**		
Coolant		5 - Compressed air	2 - External	5 - Compressed air	5 - Compressed air		
Cutting Time		240	150	100	60		
Tool Life (pcs)		1	1	1	1		
Spindle Load (%)		20	10	15	15		
Remark			외부급유 사용 마그네틱 소재고정으로 칩 소재 잔존 사용 불가	**생산성 2.3배 개선** 중삭공정 추가 사용	**생산성 4배 개선** 황삭공정 적용 완료		

Material & Fixture

Part Name	MOLD
Raw Material	14 - Pre-Machined
Group	2 - Non-alloy steel and cast steel, free cutting steel
Std. Code	8 - JIS
Material	S45C
Hardness	25
Unit	HRC
Rigidity	1 - Stable

Test Result

1. 가공성 양호
2. 소재 고정 마그네틱 사용으로 절삭유 사용 불가
3. 경쟁사 대비 최종 4배 개선 완료로 중삭 공정 추가 동시 적용 완료
4. FFX4(황삭) + MDR(중삭) 공정으로 소재 2~3개 가공 활용

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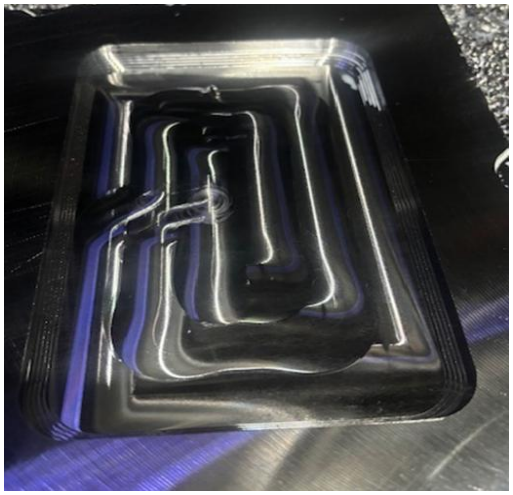
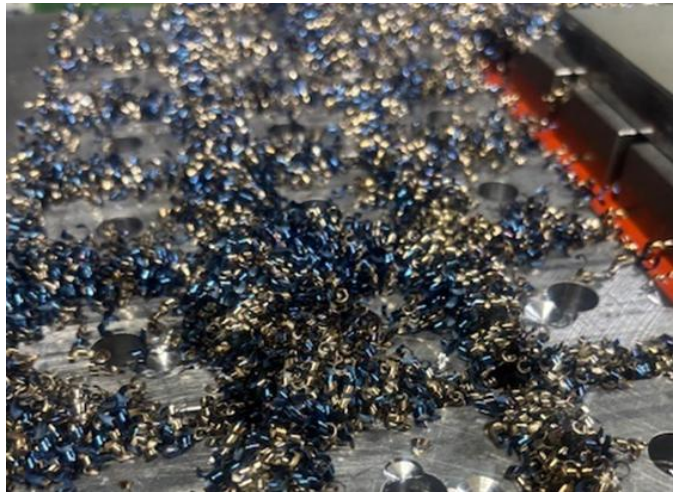
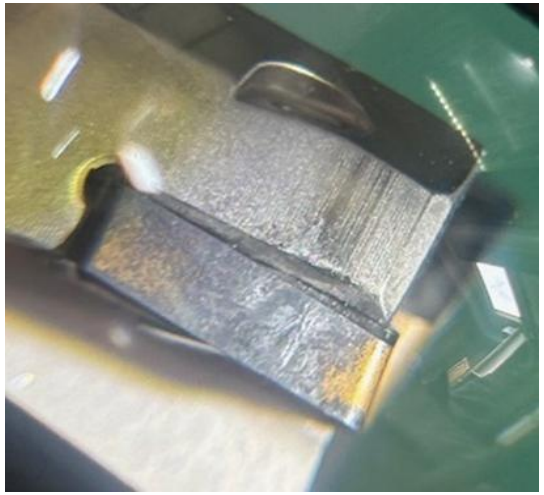
MDR D25 사용



MDR 사용





절삭유 사용 시 자석에 의한 칩 제어 불가

FFX4 D25 사용